

Ezra Satria Bagas Airlangga

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PROFILE

AI/ML Engineer with hands-on experience in multi-agent RAG systems, computer vision, deep learning, and classical ML — shipped end-to-end from data pipeline to deployed inference API. Master's student in Electrical Engineering at Telkom University (B.Eng. GPA 3.61/4.00). Stack: Python, LangChain/LangGraph, Qdrant, OpenAI, YOLO, FastAPI, Cloud Run, BigQuery.

TECHNICAL SKILLS

Languages: Python, C++, SQL/MySQL

ML / DL: scikit-learn, XGBoost, PyTorch, TensorFlow/Keras, Random Forest, SVM

LLM / RAG: LangChain, LangGraph, Qdrant, OpenAI API, Langfuse, multi-agent orchestration

Computer Vision: YOLOv12, OpenCV, vegetation indices, Vision-Language Models (Gemini, GPT-4o)

Backend / MLOps: FastAPI, Flask, Streamlit, Docker, REST API

Cloud / Data: Google Cloud Run, Cloud Build, Cloud Storage, Firebase, BigQuery, Looker Studio

IoT / Embedded: ESP32-CAM, Arduino, PlatformIO, LoRa

PROJECTS

Olist AI Assistant — E-Commerce Intelligence Platform

Apr 2026

Final Project, Purwadhika Digital School

LangGraph, GPT-4o, Qdrant, FastAPI

- Built a supervisor-based multi-agent RAG platform routing natural-language queries to semantic, SQL, and vision-search agents over Brazilian e-commerce data.
- Integrated Qdrant Cloud embeddings, GPT-4o image search, and Langfuse tracing; deployed containerized FastAPI on Cloud Run with a Streamlit + Plotly dashboard.
- github.com/papapipopepo/Olist-AI-E-Commerce-Assistance

IMDB RAG Agent — Multi-Agent Movie Chatbot

Jan 2026

RAG / LLM Project

LangChain, LangGraph, Qdrant

- Designed a supervisor-routed multi-agent RAG chatbot grounded on IMDB Top 1000 to mitigate hallucination, with source attribution and rolling chat history.
- Added token-usage and cost-estimation transparency; vectorized 1,000 documents in Qdrant.
- github.com/papapipopepo/imdb-rag-agent

SawahSmart — Rice Maturity Classification (ESP32-CAM + Cloud ML)

Jul 2025 – Present

AI & IoT Engineer, Master's Thesis

VLM, Hybrid CNN+VI, Cloud Run

- Built a 2-tier pipeline: a VLM gatekeeper (Gemini 2.5 Flash-Lite) filters non-rice images, then a Hybrid deep-learning classifier (EfficientNet CNN features fused with vegetation indices) predicts 3 growth stages.
- Ran a comparative benchmark across VLM, classical ML, deep learning (CNN), and Hybrid models: deployed Hybrid+ExGR reached test F1 0.966 (n=104); Hybrid+NGRDI led cross-validation at F1 0.977, beating the EfficientNetB0 DL baseline.
- Deployed inference API on Cloud Run with token auth; integrated ESP32-CAM, Firebase RTDB, and a PWA dashboard.
- github.com/papapipopepo/sawah-smart

Vehicle Detection & Traffic Density — YOLOv12

Feb 2026

Computer Vision Project

YOLOv12, Supervision, Streamlit

- Trained a YOLOv12 detector over 3 vehicle classes reaching mAP@50 0.902 / mAP@50-95 0.714, with LOW/MED/HIGH traffic-density classification.
- Shipped a Streamlit app for image, video, and batch inference with an analytics dashboard.
- github.com/papapipopepo/yolov12-vehicle-detection-system

Bike Sharing Demand Prediction

Dec 2025

Machine Learning Project (Regression)

XGBoost, scikit-learn

- Tuned an XGBoost regressor for hourly demand reaching R^2 0.912 (RMSE 65.4, MAE 43.0); surfaced commute-hour and seasonal error patterns. github.com/papapipopepo/Bike-Sharing-Demand-Prediction

Flood Disaster Detection Wireless Sensor Network

Jul 2023 – Aug 2023

IoT Engineer, 3-person team

IoT, ML, dashboard

- Built an IoT + ML flood-detection system with a monitoring dashboard; led device integration. Registered as an official copyright (Hak Cipta, DJKI No. 000654338).

EXPERIENCE

Big Data Analyst – Project-Based Internship <i>PT Kimia Farma, Tbk. × Rakamin Academy</i> - Scored 89.03 (Excellent); processed large-scale transactional data in BigQuery with scheduled analytics workflows. - Built an interactive Looker Studio dashboard (2020–2023) for sales, profit, ratings, and regional performance, filterable by province, city, and branch.	Apr 2025 – May 2025 <i>Bandung</i>
Virtualization Engineering – Internship <i>PT Sumber Segara Primadaya</i> Performed Proxmox/VMware installation and configuration; set up Ubuntu containers with MySQL, Apache2, PHP, and phpMyAdmin for a sample web server.	Jul 2023 – Aug 2023 <i>Cilacap</i>

EDUCATION

Telkom University <i>M.Eng. Electrical Engineering</i>	Sep 2024 – Present <i>Bandung</i>
Purwadhika Digital School <i>AI Engineering Bootcamp (Job Connector)</i>	Oct 2025 – May 2026 <i>Jakarta</i>
Telkom University <i>B.Eng. Telecommunication Engineering – GPA 3.61/4.00</i>	Sep 2020 – Aug 2024 <i>Bandung</i>

CERTIFICATIONS & CREDENTIALS

Job Connector Bootcamp: AI Engineering – Purwadhika Digital Technology School (2025–2026), No. 202510008961
Copyright / Hak Cipta, No. 000654338 – DJKI, Ministry of Law & Human Rights: AI-based Flood Detection WSN Web Application (2024)
Intro to Data Analytics – RevoU (Mar 2025)
Data Analysis Fundamental – MySkill (Apr 2025)

LANGUAGES

Bahasa Indonesia (Native), English (Intermediate)
